

Practice Questions for Portfolio Selection (Markowitz Model)

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1 Questions

1. In the Markowitz portfolio selection model, what is the primary objective of the optimization problem, and how does this objective relate to the investor's risk tolerance and expected return?
 - a. To maximize the portfolio's variance, representing the highest possible risk.
 - b. To minimize the portfolio's expected return, ensuring the lowest possible profit.
 - c. To find a portfolio that balances risk and return, optimizing a combination of variance and expected return based on the investor's preferences.
 - d. To maximize the portfolio's covariance, indicating the strongest possible correlation between assets.
 - e. To minimize the portfolio's Sharpe ratio, resulting in the lowest risk-adjusted return.
2. In the Markowitz model, how are the constraints typically formulated to reflect the limitations on the investor's budget and any restrictions on the proportion of the portfolio invested in individual assets?
 - a. The constraints are formulated as inequalities, allowing for flexibility in asset allocation but ensuring the budget is not exceeded.
 - b. The constraints are formulated as equalities, strictly defining the proportion of the portfolio invested in each asset.
 - c. The constraints are formulated as a combination of equalities and inequalities, depending on the specific asset allocation requirements.
 - d. The constraints are not explicitly formulated; the model relies solely on the objective function to determine the optimal asset allocation.
 - e. The constraints are formulated using non-linear equations to capture the complex relationships between assets.
3. In the context of the Markowitz portfolio selection model, what is the economic interpretation of the dual variables (shadow prices) associated with the constraints, and how can this information be used to improve portfolio construction?
 - a. The dual variables represent the marginal increase in portfolio variance for each additional dollar invested.
 - b. The dual variables represent the marginal increase in portfolio expected return for each additional dollar invested.

- c. The dual variables represent the marginal increase in portfolio risk for each additional unit of a specific asset added.
 - d. The dual variables represent the marginal increase in portfolio return for each additional unit of a specific asset added.
 - e. The dual variables represent the marginal change in the portfolio's Sharpe ratio for each additional dollar invested.
4. Suppose the Markowitz portfolio optimization has been solved, and the optimal solution indicates that a particular asset is not included in the portfolio (its weight is zero). What information can be obtained from the reduced costs associated with this asset, and how can this information be used to improve portfolio construction?
- a. The reduced cost indicates the marginal increase in portfolio variance if that asset were included.
 - b. The reduced cost indicates the marginal decrease in portfolio expected return if that asset were included.
 - c. The reduced cost indicates the marginal increase in portfolio expected return if that asset were included.
 - d. The reduced cost indicates the marginal change in the portfolio's Sharpe ratio if that asset were included.
 - e. The reduced cost provides no useful information about the asset's potential contribution to the portfolio.
5. In the context of the Markowitz portfolio selection model, explain how the objective function is formulated to balance the investor's desire for high expected return with their aversion to risk. How are risk and return represented mathematically in the objective function, and how does the investor's risk tolerance influence the weighting of these two factors?
- a. The objective function maximizes expected return, ignoring risk entirely.
 - b. The objective function minimizes risk, ignoring expected return entirely.
 - c. The objective function maximizes a weighted combination of expected return and the negative of portfolio variance, where the weights reflect the investor's risk aversion.
 - d. The objective function minimizes the portfolio variance subject to a minimum acceptable level of expected return.
 - e. The objective function maximizes the Sharpe ratio (excess return per unit of risk), which directly incorporates both return and risk.
6. In the Markowitz portfolio selection model, how is risk typically represented mathematically, and what is the significance of this representation in the context of portfolio optimization?
- a. Risk is represented by the variance of the portfolio's return, signifying the volatility of the investment.
 - b. Risk is represented by the expected return of the portfolio, signifying the potential profit.
 - c. Risk is represented by the covariance matrix of asset returns, signifying the correlation between assets.
 - d. Risk is represented by the sum of the individual asset returns, signifying the total return.

- e. Risk is represented by the standard deviation of the portfolio's return, signifying the average return.
7. In the Markowitz portfolio selection model, what is the typical form of the objective function, and what does it aim to optimize?
- a. Minimize the portfolio variance subject to a minimum expected return constraint.
 - b. Maximize the portfolio's expected return subject to a maximum variance constraint.
 - c. Maximize the portfolio's Sharpe ratio without any constraints.
 - d. Minimize the portfolio's Sharpe ratio subject to a budget constraint.
 - e. Maximize the portfolio's expected return without considering risk.
8. Suppose an investor has a budget of \$100,000 to invest in two assets, A and B. Asset A has an expected return of 10
- a. $x_A + x_B \geq 100000$
 - b. $x_A + x_B \leq 100000$
 - c. $0.1x_A + 0.15x_B \leq 100000$
 - d. $0.1x_A + 0.15x_B \geq 100000$
 - e. $x_A + x_B = 0$
9. In the Markowitz model, if an asset has a reduced cost of -2
- a. The asset should be included in the portfolio to increase the portfolio's return.
 - b. The asset should be excluded from the portfolio because it reduces the portfolio's return.
 - c. The asset's inclusion or exclusion has no impact on the portfolio's return.
 - d. The reduced cost is irrelevant to the asset's inclusion in the portfolio.
 - e. More information is needed to determine the asset's potential inclusion.

2 Answers

1. In the Markowitz portfolio selection model, what is the primary objective of the optimization problem, and how does this objective relate to the investor's risk tolerance and expected return?
 - a. Maximizing variance is the opposite of the Markowitz model's goal. The model typically aims to minimize variance (risk) while maximizing return.
 - b. Minimizing expected return is counterintuitive to investment goals. The model aims to maximize expected return while managing risk.
 - c. The Markowitz model aims to find an optimal portfolio that balances risk and return. The investor's risk tolerance is implicitly incorporated through the weighting given to variance (risk) versus expected return in the objective function. Different investors will have different preferences, leading to different optimal portfolios.
 - d. Maximizing covariance is not the primary objective. While covariance is a component of the portfolio's variance, the goal is to manage the overall portfolio variance, not just the covariance between assets.
 - e. Minimizing the Sharpe ratio is contrary to the model's purpose. The Sharpe ratio measures risk-adjusted return, and the model typically aims to maximize this ratio.
2. In the Markowitz model, how are the constraints typically formulated to reflect the limitations on the investor's budget and any restrictions on the proportion of the portfolio invested in individual assets?
 - a. While inequalities are used for some constraints (e.g., limits on individual asset proportions), an equality constraint is necessary to ensure the entire budget is used.
 - b. Strict equalities for all asset proportions would be overly restrictive and unrealistic in most portfolio selection scenarios.
 - c. The constraints typically include an equality constraint representing the budget constraint (the sum of investments must equal the total budget). Additional constraints might be inequalities, limiting the proportion of the portfolio invested in any single asset to avoid excessive risk associated with over-concentration in a particular asset.
 - d. The constraints are essential for defining the feasible region of the optimization problem. The objective function alone cannot determine the optimal allocation without considering the limitations on budget and asset proportions.
 - e. The Markowitz model uses linear constraints to represent the budget and asset allocation restrictions. Non-linear constraints would significantly complicate the optimization problem.
3. In the context of the Markowitz portfolio selection model, what is the economic interpretation of the dual variables (shadow prices) associated with the constraints, and how can this information be used to improve portfolio construction?
 - a. Dual variables relate to the change in the objective function (expected return), not directly to variance.
 - b. The dual variables associated with the budget constraint represent the marginal increase in expected return for each additional dollar invested. This information is valuable for understanding the potential benefit of increasing the investment budget. Dual variables associated with asset proportion constraints indicate the marginal increase in expected return from relaxing those constraints.

- c. While dual variables relate to changes in the objective function, they don't directly represent the marginal change in risk for adding a specific asset. The impact on risk is indirect through the change in the portfolio's overall variance.
 - d. Similar to option C, the dual variables don't directly represent the marginal change in return for adding a specific asset. The impact on return is indirect through the change in the portfolio's overall expected return.
 - e. The Sharpe ratio is a function of both return and risk. While dual variables can indirectly influence the Sharpe ratio, they don't directly represent its marginal change.
4. Suppose the Markowitz portfolio optimization has been solved, and the optimal solution indicates that a particular asset is not included in the portfolio (its weight is zero). What information can be obtained from the reduced costs associated with this asset, and how can this information be used to improve portfolio construction?
- a. Reduced costs relate to changes in the objective function (expected return), not directly to variance.
 - b. A negative reduced cost for an excluded asset indicates that including a small amount of that asset would decrease the portfolio's expected return. A positive reduced cost indicates that including a small amount of that asset would increase the portfolio's expected return. This information helps in evaluating whether the asset's exclusion is justified or if there's potential for improvement by including it.
 - c. A positive reduced cost indicates an increase in expected return, not a decrease.
 - d. While reduced costs can indirectly influence the Sharpe ratio, they don't directly represent its marginal change.
 - e. Reduced costs provide valuable information for evaluating the potential contribution of excluded assets.
5. In the context of the Markowitz portfolio selection model, explain how the objective function is formulated to balance the investor's desire for high expected return with their aversion to risk. How are risk and return represented mathematically in the objective function, and how does the investor's risk tolerance influence the weighting of these two factors?
- a. This is incorrect; the Markowitz model explicitly considers both risk and return. Ignoring risk would lead to a highly risky portfolio.
 - b. This is incorrect; the Markowitz model aims to maximize return while managing risk, not solely minimize risk.
 - c. The Markowitz model aims to find an optimal portfolio that balances risk and return. The objective function typically maximizes a weighted combination of expected return and the negative of portfolio variance (or standard deviation). The weights reflect the investor's risk tolerance: a higher weight on return indicates higher risk tolerance, while a higher weight on the negative of variance indicates greater risk aversion. This allows the investor to customize the portfolio to their specific risk profile.
 - d. This is a valid optimization approach (minimizing variance subject to a return constraint), but it's not the typical way the Markowitz objective function is formulated. The typical formulation is a weighted combination of return and risk.
 - e. While maximizing the Sharpe ratio is a related and often used approach, it's not the standard formulation of the Markowitz objective function. The Markowitz model typically directly incorporates expected return and variance (or standard deviation).

6. In the Markowitz portfolio selection model, how is risk typically represented mathematically, and what is the significance of this representation in the context of portfolio optimization?
- Risk is indeed represented by the variance (or standard deviation) of the portfolio's return. A higher variance indicates greater volatility and thus higher risk.
 - The expected return is a measure of the potential profit, not risk. The Markowitz model considers both return and risk.
 - The covariance matrix is used to calculate the portfolio variance, but it's not the direct representation of risk itself.
 - The sum of individual asset returns doesn't account for the correlations between assets, which are crucial for determining overall portfolio risk.
 - Standard deviation is the square root of variance and is a common measure of risk, but the variance itself is used in the model's formulation.
7. In the Markowitz portfolio selection model, what is the typical form of the objective function, and what does it aim to optimize?
- The standard Markowitz model aims to minimize portfolio variance (risk) while meeting a minimum expected return target.
 - While maximizing return is a goal, the Markowitz model explicitly incorporates risk by constraining the variance.
 - The Sharpe ratio is a useful metric, but it's not directly the objective function in the basic Markowitz model.
 - Minimizing the Sharpe ratio is counterintuitive; the goal is to maximize it.
 - Ignoring risk is unrealistic and would lead to highly volatile portfolios.
8. Suppose an investor has a budget of \$100,000 to invest in two assets, A and B. Asset A has an expected return of 10
- This constraint implies that the investor must invest at least \$100,000, which is not a limitation.
 - The total investment cannot exceed the budget, hence the inequality $x_A + x_B \leq 100000$.
 - This constraint mixes investment amounts with returns, which is incorrect.
 - Similar to A, this implies a minimum investment, not a maximum.
 - This implies no investment at all.
9. In the Markowitz model, if an asset has a reduced cost of -2
- A negative reduced cost suggests that including the asset would improve the objective function (portfolio return), so it's a candidate for inclusion.
 - A negative reduced cost indicates a potential *increase* in return, not a decrease.
 - The reduced cost directly reflects the potential change in the objective function upon including the asset.
 - The reduced cost is a key piece of information in sensitivity analysis for portfolio optimization.
 - The reduced cost provides sufficient information to assess the asset's potential contribution.